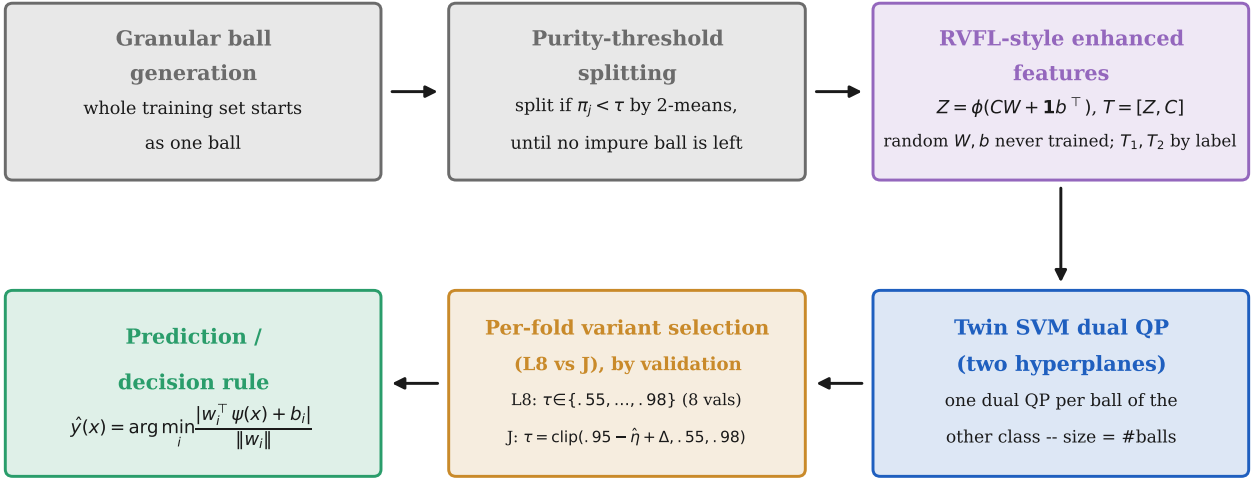
Granular ball generation
whole training set starts as one ball
Purity-threshold splitting
split if $\pi_j < \tau$ by 2-means, until no impure ball is left
RVFL-style enhanced features
$Z = \phi(CW + \mathbf{1}b^\top)$, $T = [Z, C]$
random W, b never trained; T_1, T_2 by label
Twin SVM dual QP (two hyperplanes)
one dual QP per ball of the other class -- size = #balls
Per-fold variant selection (L8 vs J), by validation
L8: $\tau \in \{.55, \ldots, .98\}$ (8 vals)
J: $\tau = \mathrm{clip}(.95 - \hat{\eta} + \Delta, .55, .98)$
Prediction / decision rule
$\hat{y}(x) = \arg\min_i \frac{|w_i^\top \psi(x) + b_i|}{\|w_i\|}$